





Relational Calculus and Codd's Rules

From “How” to “What”: Declarative Queries and Relational DBMS Principles



Learning Objectives

By the end of this lecture, you should be able to...

01

Understand Relational Calculus as a declarative, non-procedural approach based on predicate logic.

02

Differentiate Relational Algebra and Relational Calculus — HOW to retrieve versus WHAT is required.

03

Explain Tuple Relational Calculus (TRC) and how tuple variables represent complete records.

04

Explain Domain Relational Calculus (DRC) and how domain variables represent attribute values.

05

Understand Codd's Rules and why relational principles remain important in modern database systems.



Revision: Relational Algebra

RECALL

From the previous lectures: a procedural way to express a query

RELATIONAL ALGEBRA

Example: Find names of CSE students

$\pi_{\text{Name}} (\sigma_{\text{Dept} = \text{'CSE'}} (\text{Student}))$

- Select the tuples that satisfy the condition.
- Project only the attribute needed in the answer.
- The expression describes a sequence of operations.

The key question is...

“HOW do I obtain the result?”

Operations tell the DBMS
the route to the answer.



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Introduction to Relational Calculus

CORE IDEA

A declarative, predicate-logic view of querying

RELATIONAL CALCULUS

“What information is required?”

- Non-procedural / declarative
- Based on mathematical predicate logic
- Conditions describe the required result
- The DBMS determines how to retrieve it

Same question, different viewpoint

Find students from CSE

Condition: Department = CSE

System chooses the retrieval strategy



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Relational Algebra vs Relational Calculus

COMPARE

Two formal perspectives on the same relational query

Dimension	Relational Algebra	Relational Calculus
Style	Procedural	Non-procedural
Focus	How to obtain the result	What result is required
Core language	Operations	Predicates / logic

Algebra

SELECT → FILTER → PROJECT

Calculus

“Return tuples satisfying P.”



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Types of Relational Calculus

TWO FORMS

Two forms : the difference is what the variable represents

1 Tuple Relational Calculus (TRC)

Variable → complete tuple / row

Think: “Which rows satisfy the predicate?”

2 Domain Relational Calculus (DRC)

Variable → individual attribute value

Think: “Which values satisfy the predicate?”

Memory cue: T = Tuple = Table Row | D = Domain = Data Value



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Tuple Relational Calculus (TRC)

TRC

Variables represent tuples — complete rows of a relation

General form

$$\{ t \mid P(t) \}$$

Read it as:

**“Find all tuples t
such that $P(t)$ is true.”**

Example: CSE students

$$\{ t \mid \text{Student}(t) \wedge t.\text{Department} = \text{'CSE'} \}$$

$t \rightarrow$ tuple variable
 $P(t) \rightarrow$ predicate / condition

**The expression states the condition, not
the search procedure.**



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TRC Example: Read the Predicate, Then Read the Result

WORKED
EXAMPLE

A tiny relation makes the logic visible

RollNo	Name	Department
101	Rahul	CSE
102	Anita	ECE
103	Raj	CSE

Only tuples satisfying the predicate are returned.

TRC

$\{ t \mid \text{Student}(t) \wedge t.\text{Department} = \text{'CSE'} \}$

RollNo	Name
101	Rahul
103	Raj

No search algorithm is specified



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Domain Relational Calculus (DRC)

DRC

Variables represent attribute values rather than complete tuples

General form

$$\{ \langle x_1, x_2, \dots, x_n \rangle \mid P(x_1, x_2, \dots, x_n) \}$$

$x_i \rightarrow$ domain / attribute value
 $P \rightarrow$ logical condition

Example: return CSE names

$$\{ \langle \text{Name} \rangle \mid \exists \text{RollNo, Dept} (\text{Student}(\text{RollNo, Name, Dept}) \wedge \text{Dept} = \text{'CSE'}) \}$$

Focus: the individual values that form
the answer



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TRC vs DRC

COMPARE

The same declarative idea different variable granularity

	TRC	DRC
Variable represents	Tuple / row	Domain / attribute value
Works with	Complete records	Individual values
Mental picture	A whole row	Selected cells / values
Memory cue	T = Tuple = Table Row	D = Domain = Data Value

TRC → think ROW

DRC → think VALUE

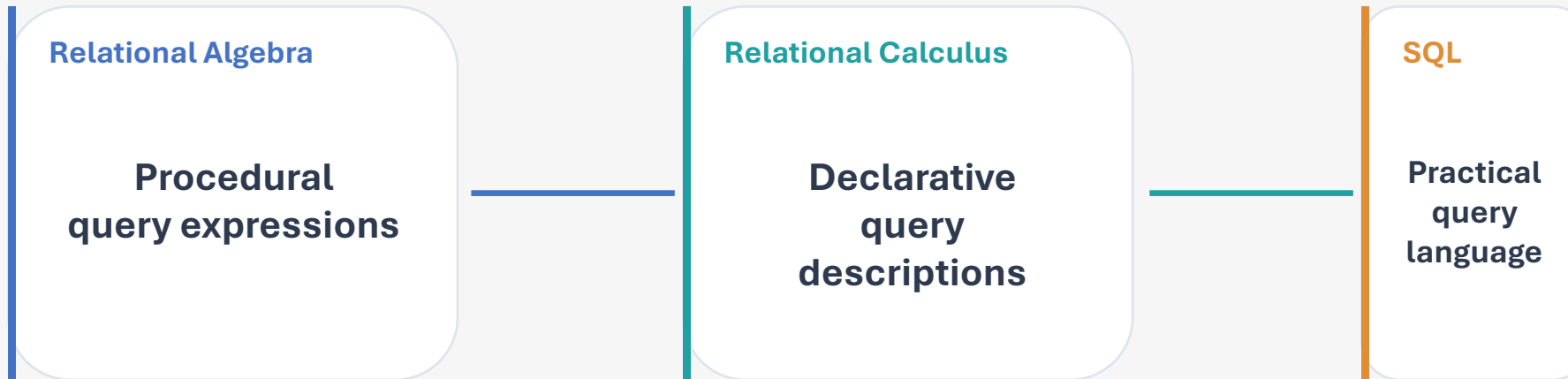


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Relational Completeness

THEORY →
PRACTICE

Can the language express every query expressible in relational algebra?



Relationally complete = able to express all queries that can be expressed using relational algebra.

Key idea: different formalisms can describe equivalent relational queries.

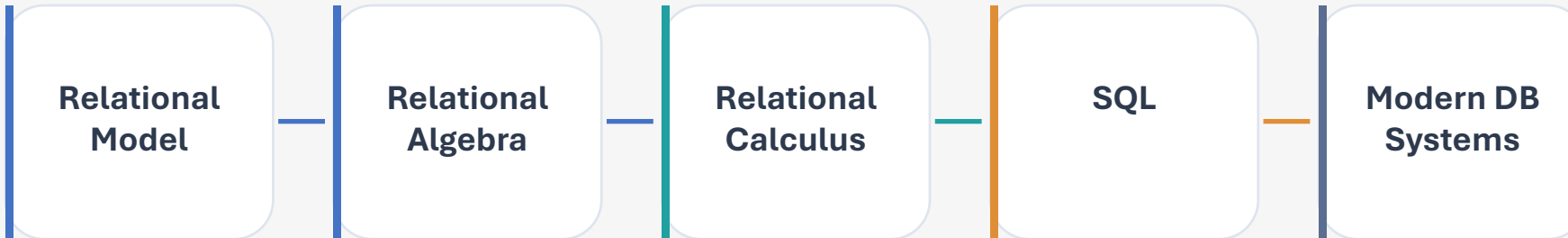


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From Relational Theory to SQL

BRIDGE

A conceptual bridge from formal foundations to the language students will use



SQL example

```
SELECT Name FROM Student WHERE Department='CSE';
```

User specifies **WHAT** is needed → DBMS decides **HOW** to retrieve it.



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Introduction to Codd's Rules

CODD

Principles for evaluating what a “true” relational DBMS should provide

Why rules?

Relational DBMS



Principles / criteria



Can we call it relational?

Codd's framework

Rule 0

Foundation rule

Rules 1–12

Additional relational principles

13 rules in total, including Rule 0



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Rule 0: Foundation Rule

RULE 0

The system must genuinely operate through relational capabilities

Rule

A system claiming to be a relational DBMS must manage databases entirely through relational capabilities.

Tables

Relations

Operations

**Relational
operations**

Foundation

**Relational
mechanism**



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Rule 1: Information Rule

RULE 1

Everything the database knows should be represented as values in tables

Rule

All information in a relational database must be represented only as values in tables.

RollNo	Name	Department
101	Rahul	CSE
102	Anita	ECE

Mental model

Information



Rows + Columns



Values



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Rule 2: Guaranteed Access Rule

RULE 2

Every individual value should have a clear relational “address”

Rule

Every individual data value must be accessible by table name + primary key value + attribute name.

Table: Student	RollNo	Email
	101	rahul@example
	102	anita@example

TABLE

KEY

ATTRIBUTE

→ uniquely points to the desired value



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Rule 3: Systematic Treatment of NULL Values

RULE 3

Missing, unknown, or not-applicable information must be handled consistently

Rule

NULL is a special representation of missing/unknown/not-applicable information — not an ordinary value such as 0.

RollNo	Name	Phone
101	Rahul	9876...
102	Anita	NULL

Do not confuse

NULL $\neq$ 0
NULL $\neq$ empty string
NULL $\neq$ “known value”



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Rule 4: Dynamic Online Catalog

RULE 4

The database should also describe itself through the relational model

Database data

**STUDENT
DEPARTMENT
ENROLLMENT**

Metadata / catalog

**Table names
Column names
Constraints
User permissions**

**Modern equivalent: data dictionary /
catalog**



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Rule 5: Comprehensive Data Sublanguage

RULE 5

A relational system needs one complete language for the database lifecycle

Rule

At least one complete language should support definition, manipulation, constraints, authorization, and transactions.

DDL

Define

DML

Manipulate

DCL

Authorize

TCL

Transactions

Constraints cut across the lifecycle: the language should support them too.



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Rules 6–12: The Remaining Principles

RULES 6–12

Do not memorize the labels in isolation, attach each rule to its design question

6 View Updating

Can logically update views?

7 High-Level INSERT / UPDATE / DELETE

Can operations work on sets?

8 Physical Data Independence

Can storage change without affecting users?

9 Logical Data Independence

Can schema change without breaking applications?

10 Integrity Independence

Are constraints independent of applications?

11 Distribution Independence

Can users ignore physical data location?

12 Non-Subversion

Can low-level access bypass integrity?



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Why Codd's Rules Still Matter

PERSPECTIVE

The rules are principles of relational design, not merely historical trivia

Relational systems

PostgreSQL
Oracle Database
MySQL
SQL Server

Many relational principles remain visible in modern DBMS design.

NoSQL perspective

Flexibility
Scalability
Distributed processing

May not follow every relational rule.

The enduring question

How should data be represented, accessed, protected, and kept independent from implementation details?



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Summary

A retrieval map for the entire lecture

RECAP

Relational Calculus

Declarative: describe WHAT is required.

TRC

Variables represent tuples / rows.

DRC

Variables represent domain / attribute values.

Completeness

Algebra and calculus can express the same relational query class.

SQL connection

SQL provides a practical declarative query language.

Codd's Rules

Principles behind a true relational DBMS.

One sentence to remember: “Algebra tells HOW; calculus tells WHAT.”



1. Explain the difference between procedural and declarative query languages.
2. Define Relational Completeness.
3. What is the Information Rule (Codd's Rule 1)?
4. Why is NULL handling treated explicitly in Codd's Rule 3?



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Question

Explain relational calculus as a declarative approach to database querying, and distinguish Tuple Relational Calculus (TRC) from Domain Relational Calculus (DRC) with suitable examples. Describe the purpose of Codd's Rules and explain how they express the principles of a relational DBMS.



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Thanks!



Heads Up!

These slides are just a **guide** — not the full story!

To really learn , make sure to read your book, and explore other resources too.